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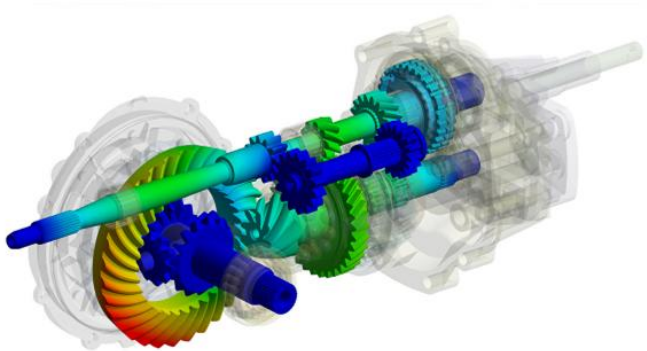
# FEA Foundations with ANSYS

Mechanical Incubator

# Why Learn FEA?

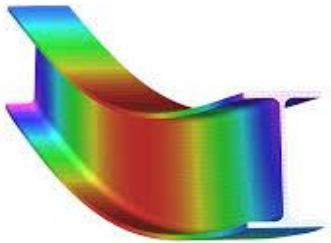
**Why Engineers Use Simulation**

- Reduce physical prototypes
- Find weak points early
- Compare multiple designs
- Improve safety
- Save time and money



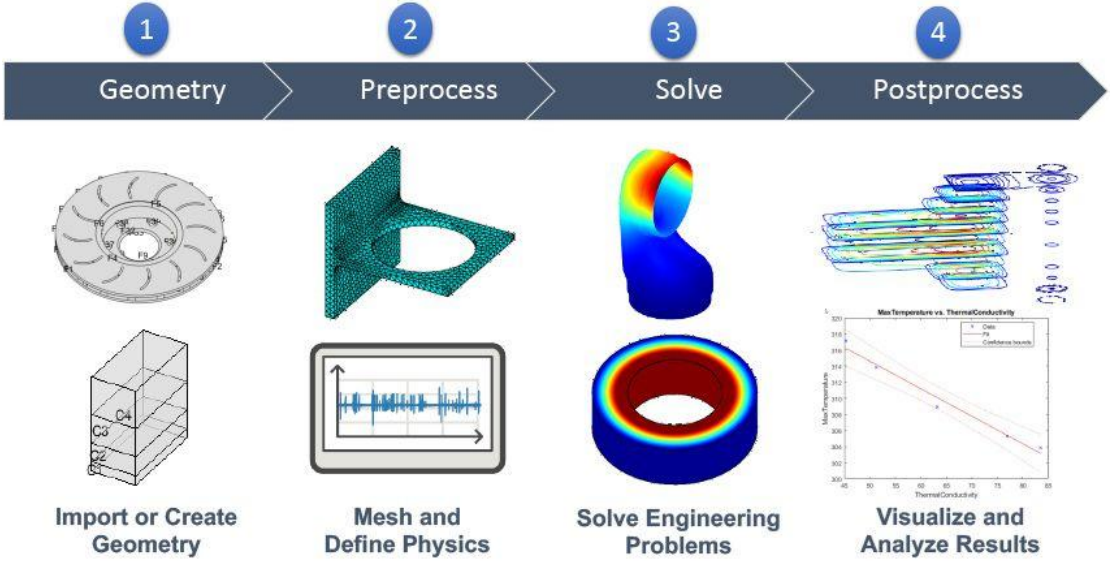
**Used in**

- Aerospace
- Automotive
- Biomedical
- Robotics
- Manufacturing



# Today's Agenda:

- **What is FEA?**
- Why engineers use it
- **ANSYS Workbench**
- Basic FEA workflow
- **First simulation**
- Reading results
- Common mistakes



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# Understanding FEA

Starting with the basics before the software

# What is Computer-Aided Engineering (CAE) ?



CAD → CAE → Manufacturing

### CAE

CAE is the broad category of software used to predict how designs behave before they are built.

It helps engineers study strength, stiffness, temperature, vibration, and performance using digital models.

### Examples of CAE

- Structural analysis
- Thermal analysis
- CFD
- Motion simulation
- FEA

### Where FEA fits

FEA is one major CAE method.

- CAD creates the geometry.
- CAE evaluates the design.
- FEA estimates how the part responds to loads and constraints.

Design → Analyze → Improve

# What is Finite Element Analysis (FEA)?

**Real part** › **Mesh it** › **Solve Physics** › **Predict Behavior**

### Typical Output Examples

- Total deformation
- Equivalent (von Mises) stress
- Strain
- Reaction forces
- Temperature / frequency / buckling

### When it helps most

When geometry, support, or loading is too messy for a quick hand calculation

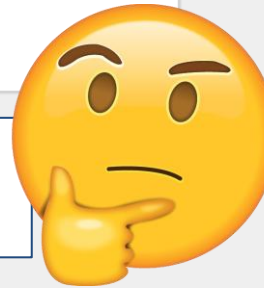
**But** you still need engineering insight before building hardware.

### Key mindset

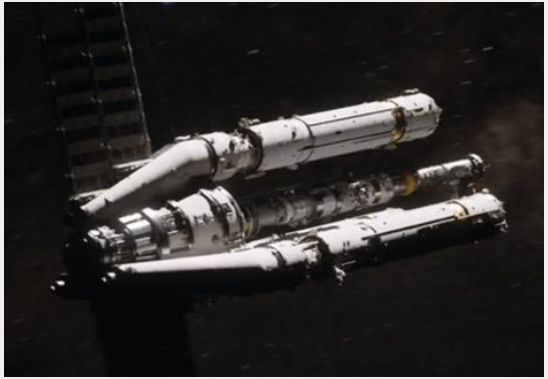
The software is fast. The thinking still matters.

A simulation is only as good as the assumptions used to build it.

A numerical method that predicts how a part responds to loads.



# FEA in Simple Terms



**Think of a LEGO Model**

Large object

↓

Many small blocks

↓

Each block is easy to analyze

↓

Together they predict the whole structure




**FEA works the same way using tiny mathematical elements.**

# Why engineers use FEA



**Find weak spots**

See where stress concentrates before something fails.

**Check stiffness**

Estimate how much a part bends or deflects under load.

**Compare concepts**

Test multiple design ideas before building all of them.

**Reduce prototyping**

Use simulation to screen bad ideas earlier.

**Optimize**

Adjust dimension or material choices

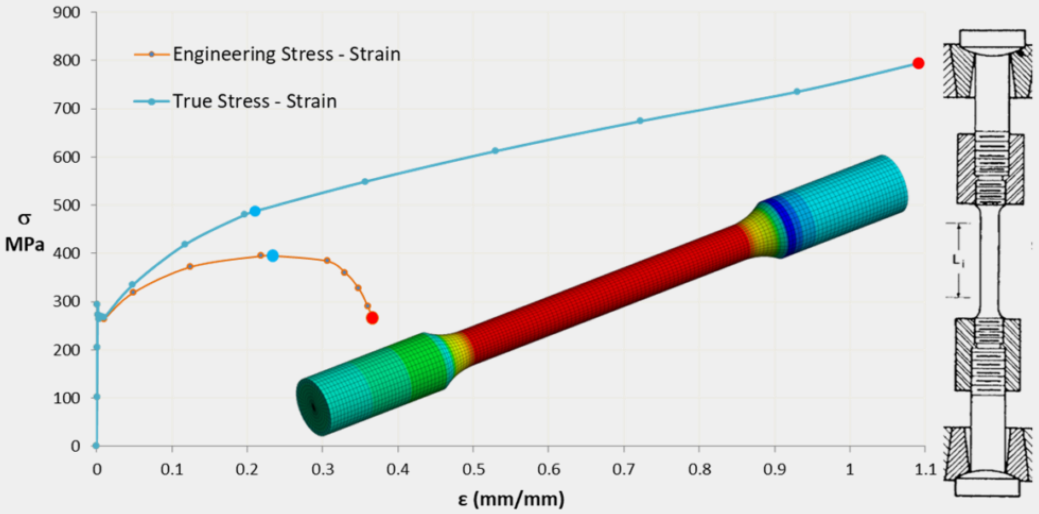
**FEA is most valuable when it supports decisions — not when it just produces colorful plots.**

- Use it to identify trends, compare alternatives, and flag areas that deserve more attention.
- Use theory and engineering judgment to decide whether the setup makes physical sense.
- Best practice is usually some combination of hand checks, simulation, and real testing.

# What kinds of questions can FEA answer?

Can it...

- Bend too much?
- Yield?
- Buckle?
- Vibrate?
- Overheat?
- Last millions of cycles?




Different analyses answer different questions.

Each questions uses different physics, but they all follow the same workflow: Build the model, apply conditions, solve, then interpret results carefully.

# FEA is powerful, But Not Magic

**What FEA does well:**

- Handles geometry too messy for a quick closed-form solution
- Shows the big picture instead of a singular value
- Makes “what-if” design changes easier to compare
- Scales to large, repetitive, and coupled engineering problems

**What can cause analysis to go wrong:**

- Wrong material properties
- Unrealistic loads or supports
- Units mistakes
- Mesh being to coarse in important regions
- Believing the plot without asking whether it matches physics

**Garbage in → garbage out still applies.**

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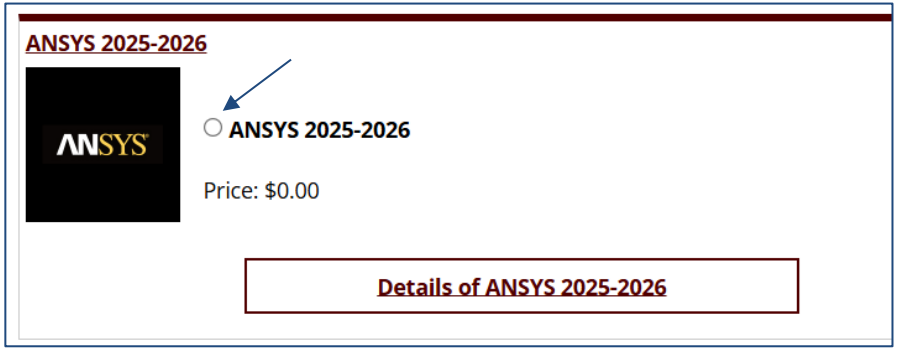
# Ansys Installation Process (Skimmed over, Refer Back to)

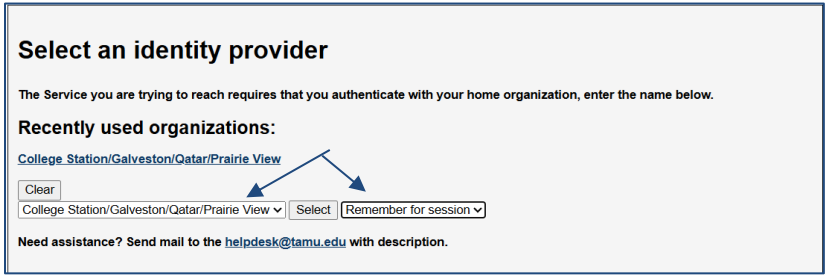
# Relatively in-depth Installation Process:

1. Available for free at tamu software center site through browser or via https://software.tamu.edu/secure/SelectProducts.aspx
2. Sign in with your Tamu NETID and Password.
3. Locate the 2025-2026 Version of ANSYS FEA
4. Send the software to your shopping cart

   by clicking the circular icon

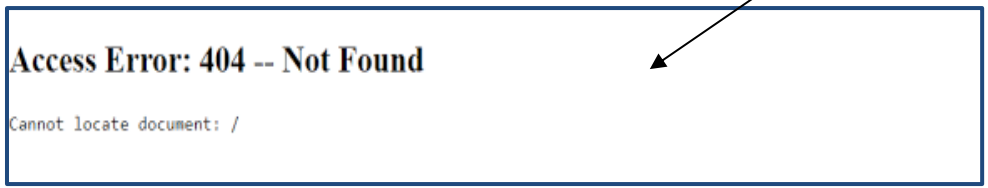
1. Fill out your identity provider, making sure it's set to “Remember for session”
2. Check out through the shopping cart and wait for a follow up email.



# Relatively in-depth Installation Process:

7. The follow up email will take you here: https://tamuengr.atlassian.net/wiki/spaces/helpdesk/pages/2147352577/Student+ANSYS+Installation+Instructions+-+Personal+Machine+Use+Only which will lead you step by step to setting up and finalizing the software.




Common Errors / Troubleshooting:

- **Make sure Your connected to tamu wifi either directly or through tamu.**
  - Instructions to download VPN can be found here: https://service.tamu.edu/TDClient/36/Portal/KB/ArticleDet?ID=346
- If you get the “Access 404 error”:
  - Ensure you are running the exe as administrator
  - Open PowerShell and run “netstat -aon”
  - Open Task Manager and the “Services” tab and end task on conflicting PID

**Good Tip: Save Often!!**

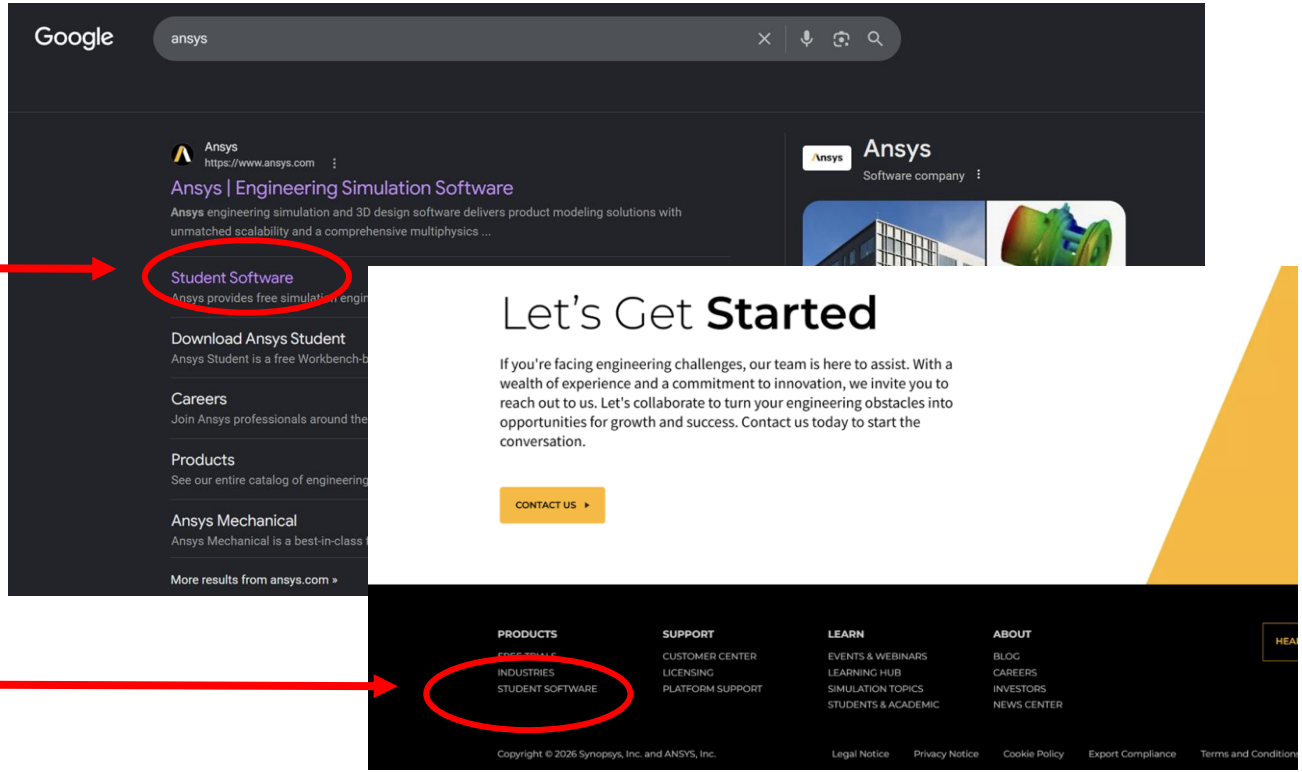
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# Alternative method to download ANSYS

(does not require account or VPN)

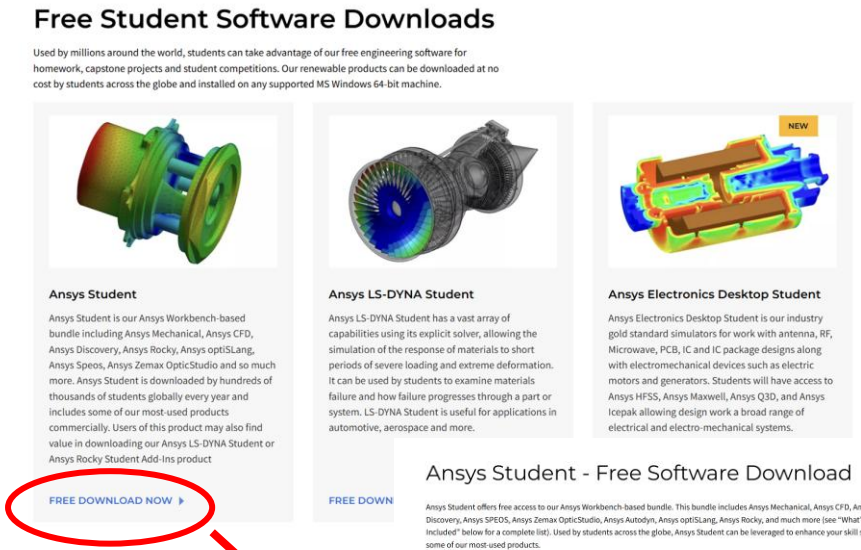
# Alternative Method

Go to the ANSYS website and click on student software

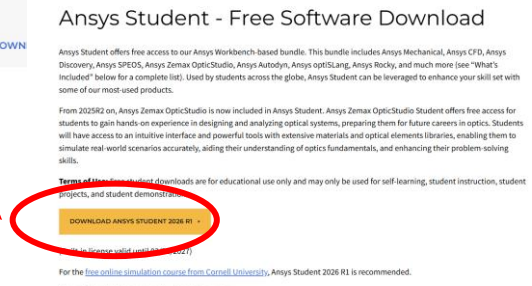


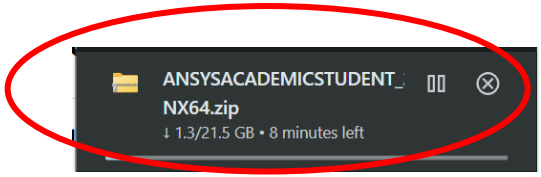
# Alternative Method

Download
ANSYS Student




This will lead to this page where you can click download

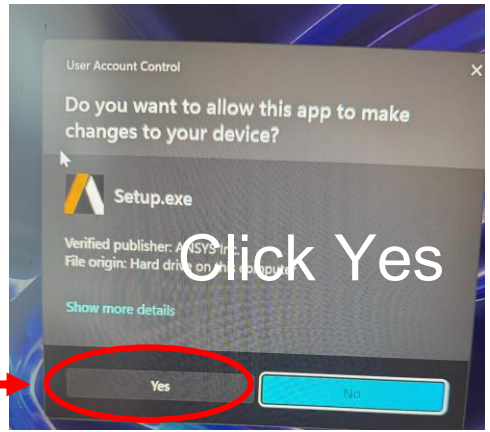


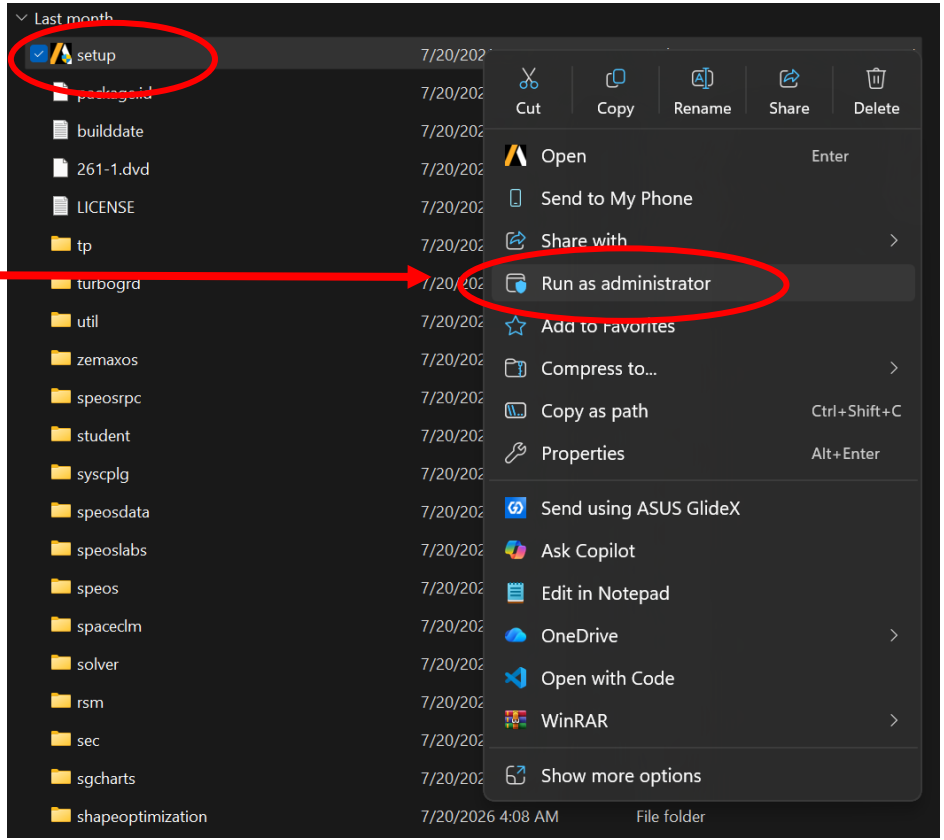

Once it is finished downloading, Unzip this file

# Alternative Method

Right click the setup exe file and run as administrator.

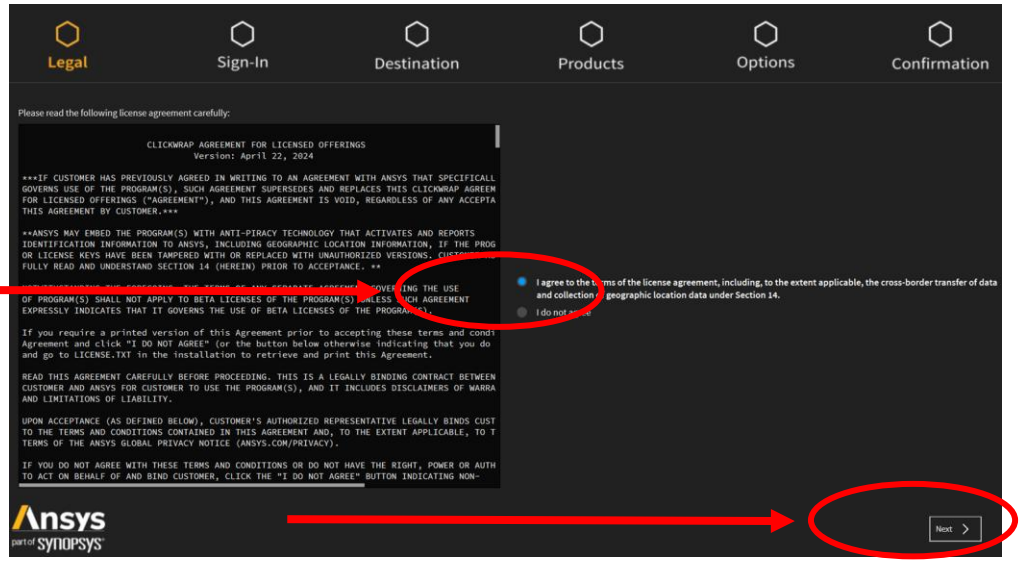


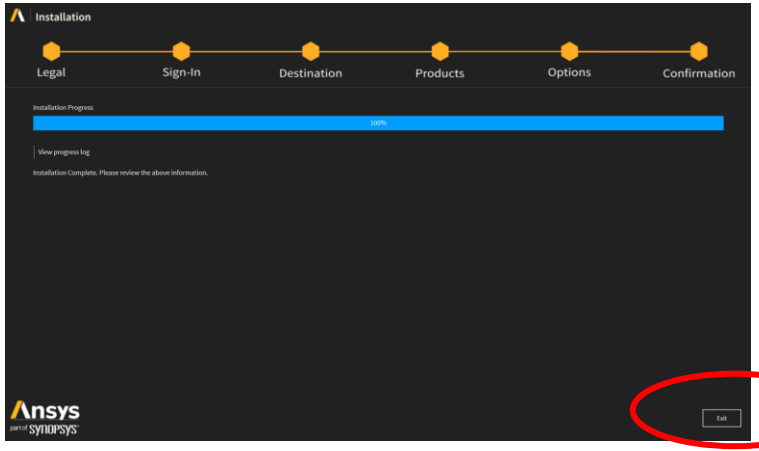

This part isnt screenshottable



# Alternative Method

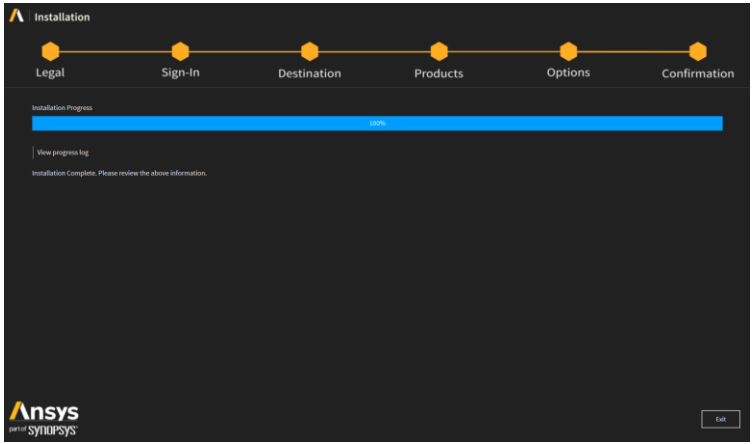
Agree to this then press next until the end



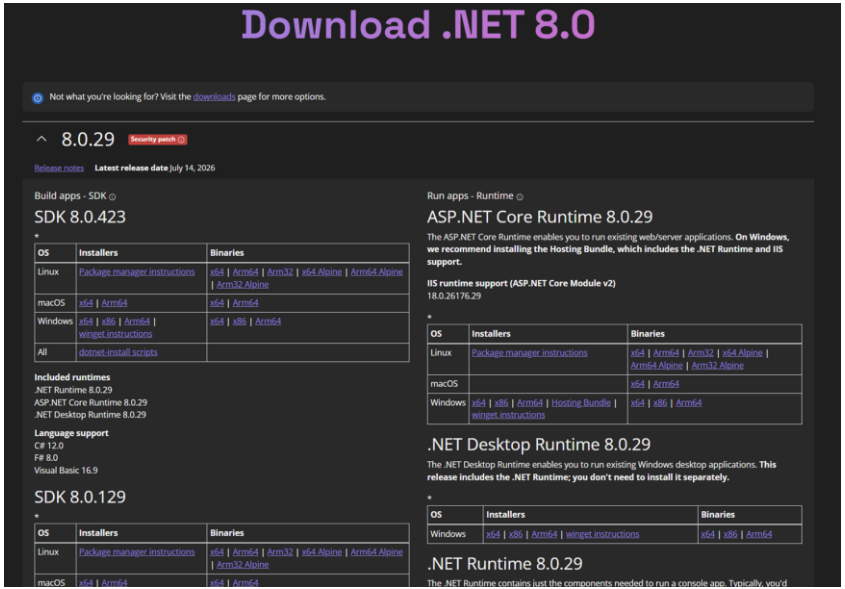




Exit and restart PC when done

# Alternative Method Extra




Install screen mightttttt prompt you to download dot Net 8.0 software

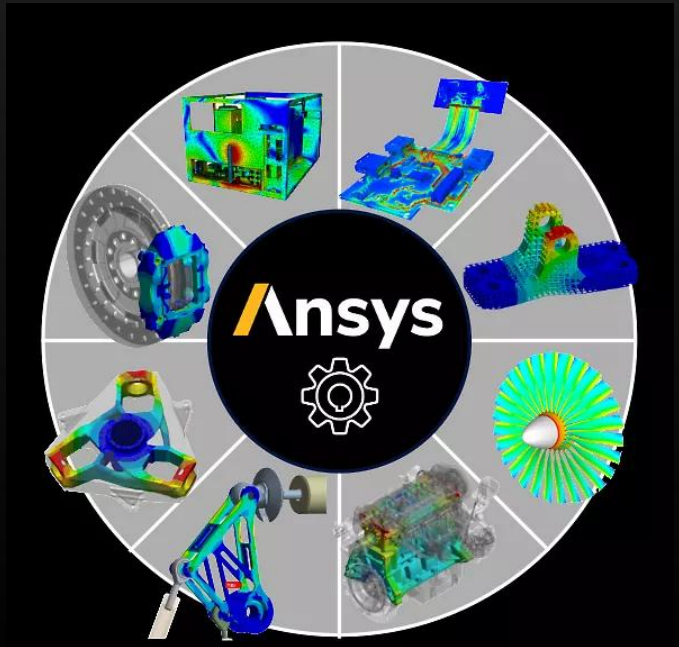



Just check what version and type its asking for and go to the dot net website and download and install that too
(outside the scope of the tutorial)

https://dotnet.microsoft.com/en-us/download/dotnet/8.0

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# Introduction to ANSYS



# Ansys ecosystem is split into:

**Workbench**

The main project environment. You connect geometry, materials, model setup, solution, and results.

**Mechanical**

The main structural simulation app. Most beginner FEA work happens here.

**Engineering Data**

Material properties live here. Even simple studies depend on good inputs.

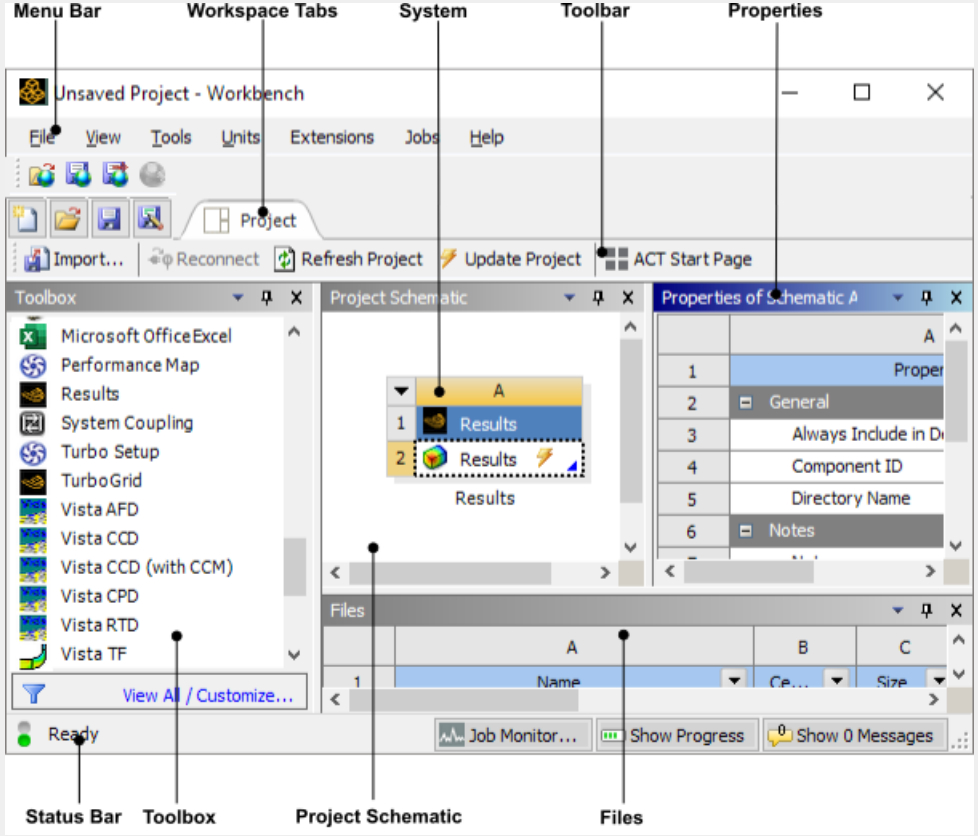
**Meshing**

The part gets broken into finite elements here. Element size strongly affects accuracy and solve time.

**Other tools**

Fluent, CFX, Discovery, SpaceClaim, and more also exist — but today focuses on Workbench + Mechanical.

# Ansys Workbench interface overview




## Toolbox

Where analysis systems live: Static Structural, Modal, Thermal, Fluent, and more.

## Project Schematic

The workflow view. Cells show Geometry → Model → Setup → Solution → Results.

## Properties / Messages

These panels explain status, missing links, and update requirements.

## How to think about it

Workbench is the organizer. Mechanical is where you actually build and solve most beginner structural models.

# Think of a project file as a *workflow*

**01**
**Drop in system**
Start with a Static Structural system in Workbench.

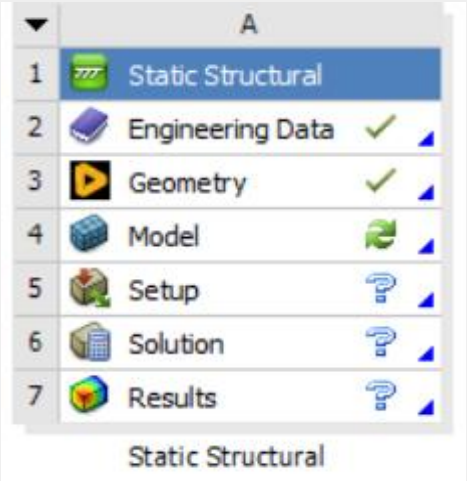
**02**
**Add geometry**
Import or create the part you want to analyze.

**03**
**Set materials**
Make sure the model has the right material properties.

**04**
**Open Mechanical**
Apply mesh, supports, loads, and requested results.

**05**
**Solve**
Let the solver calculate the response.

**06**
**Interpret**
Check whether the outputs make physical sense.




**The software flow is straightforward. Most mistakes come from assumptions, not button-clicking.**



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# The Basic FEA Workflow

The six steps you want to remember

# The beginner workflow in one picture



**01**
**Geometry**
What are we analyzing?

**02**
**Material**
What is it made of?

**03**
**Mesh**
How do we divide the model?

**04**
**Loads + Supports**
What happens to it?

**05**
**Solve**
Calculate the response

**06**
**Results**
What does it mean?

**If you remember nothing else, remember this chain.**

# Step 1: Geometry



**Goal:**

*Simple enough to solve & detailed enough to answer the question.*

**Start with the part you want to analyze**

- Import CAD
- Check dimensions & units
- Remove unnecessary details
- Keep important geometry

**Good beginner habits**

- Remove tiny fillets if they do not matter
- Suppress needless details
- Check units on import
- Make sure the body is actually connected correctly

**Ask yourself**

- What dimension really drives stiffness?
- Where would stress logically concentrate?
- Can I simplify the model and still answer the question?

# Step 2: Materials

For a basic structural analysis:

- **Young's Modulus (E)** → Stiffness
- **Poisson's Ratio (v)** → Lateral deformation
- **Density (ρ)** → Mass
- **Yield Strength** → When yielding begins

## Why this matters

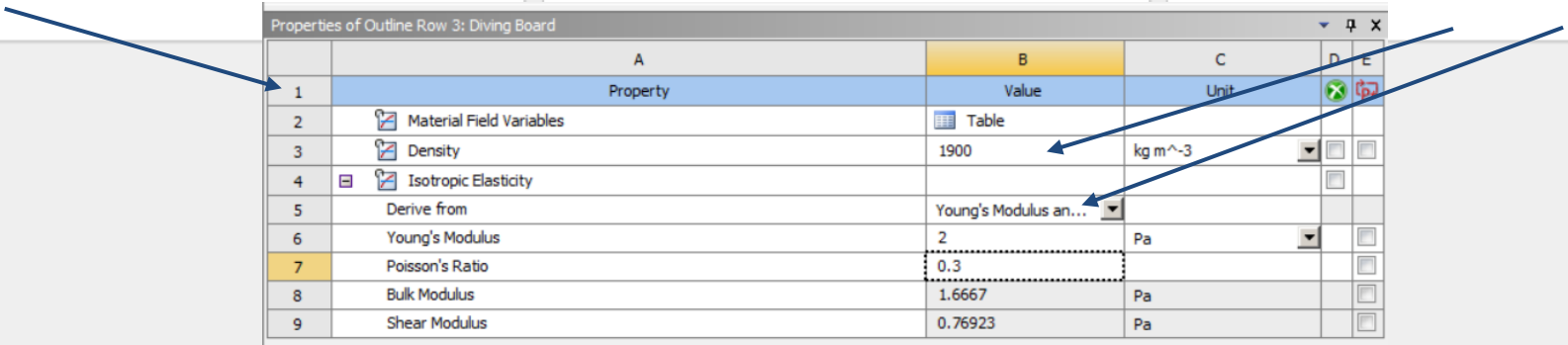
A plot can look beautiful and still be meaningless if the part is assigned the wrong material.

Young's (or elastic) modulus matters a lot because it **strongly** affects stiffness and deflection.

## Keep it realistic

If you do not know every advanced material parameter yet, that is okay.

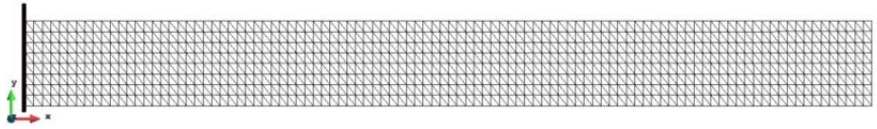
Focus on essential properties, crucial to material properties



# Step 3: Meshing

**Mesh:** a collection of small elements used to approximate the part.

More elements = More detail = More solve time



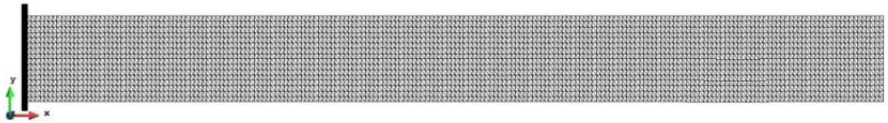


Figure 10: Finite element mesh of cantilever beam model. Mesh $B$

**Coarse Mesh**

- Fewer elements
- Faster solve
- Less detail

**Fine Mesh**

- More elements
- More detail
- Longer solve

**Mesh where it matters**

Regions near holes, fillets, contacts, and fixed supports often need more attention than large quiet regions.

**Goal:** Do not chase “tiny elements everywhere.” Chase a mesh that is refined enough to make key results stop changing much.

# Step 4: Loads and Supports

## Supports

**How is the part held?**

- Fixed Support
- Displacement
- Frictionless Support

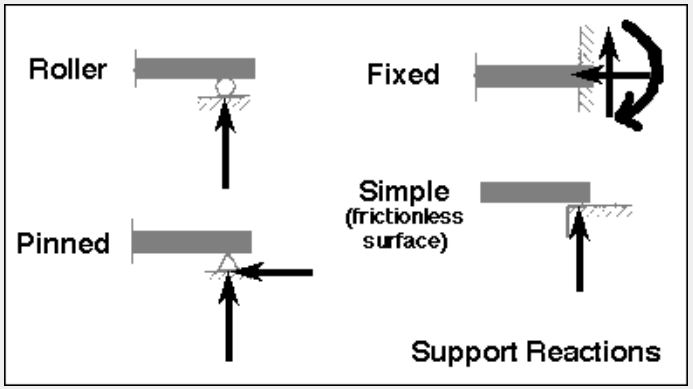
## Loads

**What acts on the part?**

- Force
- Pressure
- Moment

## Quick Tip

A model that is “locked down” in an unrealistic way can look safe in simulation while being misleading in real life.




**Model the real situation, not the easiest situation.**

# Step 5: Solve

**Before clicking Solve:**

✓ Geometry
✓ Materials
✓ Mesh
✓ Loads
✓ Supports

**After clicking Solve:**

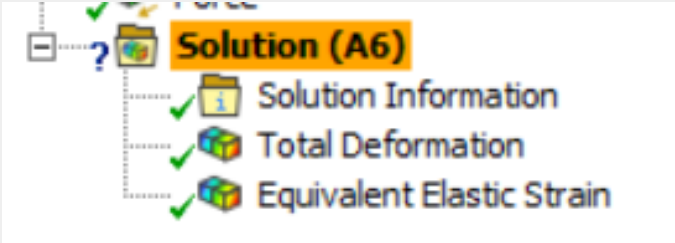
- Check warnings
- Check errors
- Inspect results

**Solved ≠ Correct**

**Quick Tip**

A successful solve is not the same thing as a correct solve.

The result still has to make engineering sense.



# Step 6: Results

ANSYS can show us:

**Deformation**

How much did it move?

Useful for stiffness and overall behavior. Remember that displayed deformation is often visually exaggerated.

**Stress**

Where is the material highly loaded?

Equivalent stress is common for judging yield risk in ductile materials.

**Strain**

How much did the material deform locally?

**Reaction Forces**

How are loads transferred through the supports?

**Keep in mind:**

Do the values, locations, and trends line up with your physical intuition?

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# Understanding Simulation Results

The plots are only useful if you know what they mean

# Deformation, Stress and Strain

**Deformation**

How far did the structure move?

Used to evaluate: stiffness, deflection, clearance and overall behaviour.

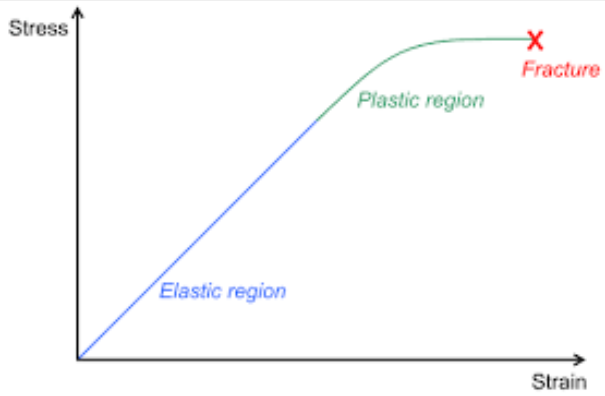
**Stress**

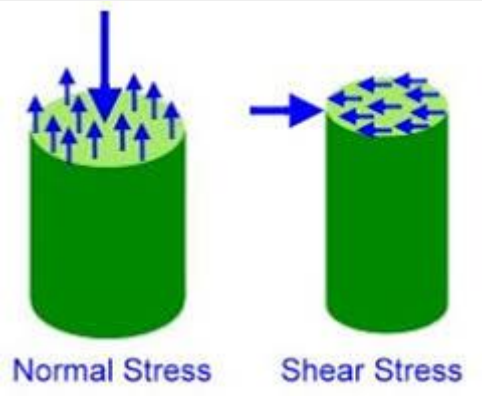
Internal force intensity within a material.

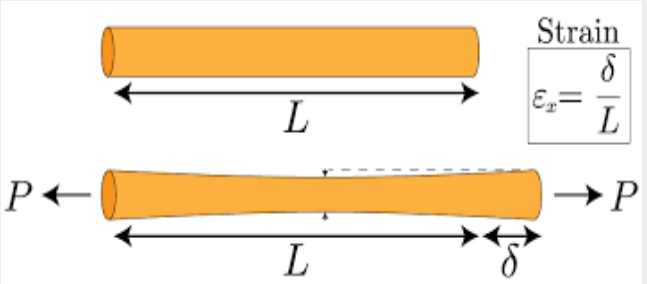
Helps indicate critical regions, potential yielding and areas needing redesign

**Strain**

How much the material terms relative to its original size.



# Von Mises Graphs

Von Mises stress are a way to represent an arbitrary three-dimensional stress state using a single positive stress value, and it is commonly used in predicting yielding of ductile materials.

This makes it a common ANSYS result.

**Important to Remember:**

If ANSYS displays a von Mises, that does not always mean failure. The correct failure criterion depends on the material and failure mode we're studying. You still need good engineering judgement to make the proper analysis.

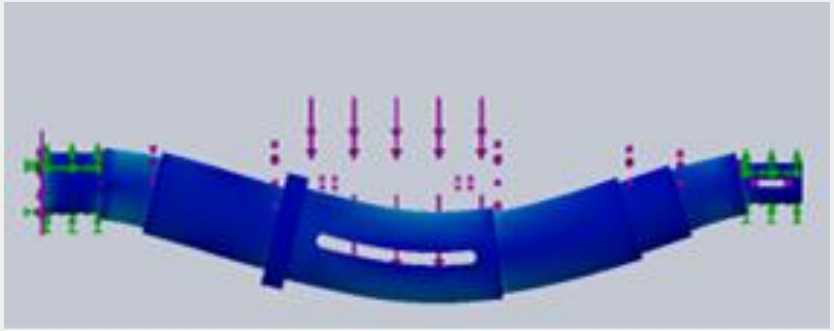
**Example interpretation:** A simulation produced a **von Mises stress of 290 MPa**. For a material with **250 MPa yield strength**, this exceeds yield, so the part **fails the first-pass yielding check**. This means the geometry, loading assumptions, or material selection should be changed.

# Exaggerated Deformation on ANSYS

**ANSYS may exaggerate the displayed deformation.**

Always check the **actual value**.

Real deflections can be tiny. To make the mode or displacement pattern visible, ANSYS often scales the shape change on screen.



Over Exaggerated Deformation ≈ 50mm
Real Deformation ≈ 4.5mm

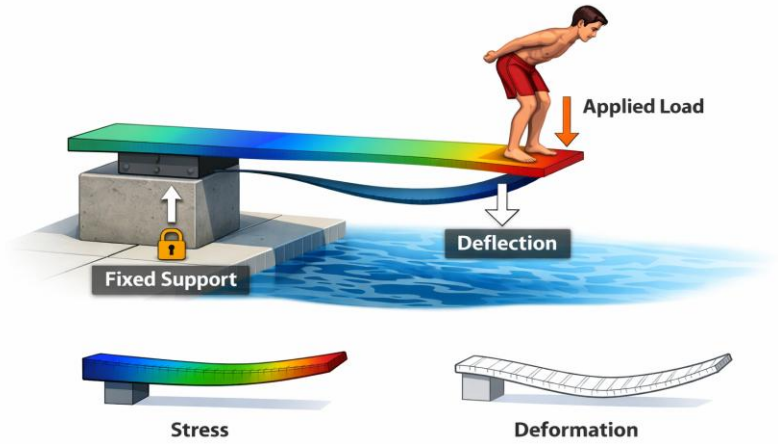
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# Example Problem!!!! (Follow along if Ansys is Installed)

Using Ansys FEA to solve the following:

# Using Ansys to Analyze A Cantilever Beam:

Problem at Hand: A local pool had one diving board that everyone claimed was “definitely fine,” even though it had started to look a little more flexible every summer. Things got more concerning when the board had to survive not just normal dives, but also the occasional overly confident cannonball attempt from someone who clearly underestimated both gravity and their own body weight. To avoid finding out its true limit the hard way, the diving board was modeled as a cantilever beam in FEA to study its deflection and stress under loading.




Assumed information:

- Person applying load is presumed to be 100kg

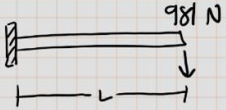
# Board Problem Solved by Hand

**Step 1. Convert person's mass into force**

$m = 100\ kg$ Weight $(W) = mg$ $g = 9.81\ m/s^2$

$W = (100)(9.81) = \boxed{981\ N}$

**Step 2. Calculate the reaction at the wall**




Vertical Reaction $(R) = W = 981\ N$

The applied load also creates a moment about the fixed support: $M_{max} = WL$

$M_{max} = 981 \cdot L\ (N \cdot m)$

**Step 3. Calculate the cross-sectional moment of inertia**

Rectangular cross section MOI: $I = \frac{bh^3}{12}$

$b$ = board width

$h$ = board thickness

**Step 4. Calculate maximum deflection**

$\delta_{max} = \frac{WL^3}{3EI}$ → Simplifies to $\boxed{\delta_{max} = \frac{WL^3}{3E\left(\frac{bh^3}{12}\right)}}$

**Step 5. Calculate maximum bending stress**

$\sigma = \frac{My}{I}$ (stress) → $\sigma_{max} = \frac{M_{max}\, y_{max}}{I}$

$M_{max} = WL$ $y_{max} = \frac{h}{2}$

$\therefore \sigma_{max} = \frac{(WL)\left(\frac{h}{2}\right)}{(bh^3)/12} = \boxed{\frac{6WL}{bh^2}}$

**Step 6. Plug numbers in and check w/ Ansys**

# Ansys workflow for the cantilever model



01

**Create beam**

Use a simple rectangular or plate-like geometry.

02

**Assign material**

Use a common engineering material with clear E and v.

03

**Fix one end**

Represent the clamped support.

04

**Apply load**

Add a downward force near the free end.

05

**Mesh**

Generate elements and refine if needed.

06

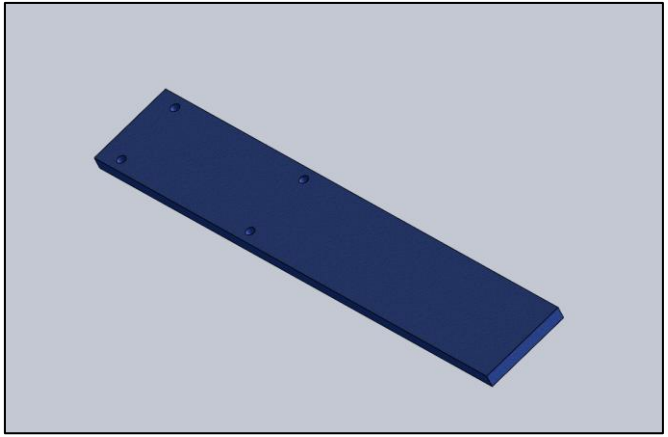
**Solve / read**

Check deformation and stress trends.

# Step 1: Stop, Think, and Plan

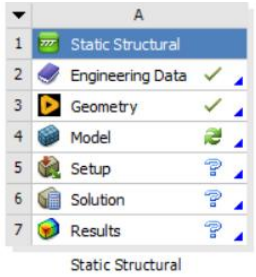
1. **Geometry:** Build the part correctly in SolidWorks, Fusion 360, or another CAD tool. For this example, the model is already created.
2. **Material:** Assign realistic material properties. In this case, an aluminum alloy is used.
3. **Supports and Loads:** Model the board with one fixed end and a free end where loading will be applied.
4. **Mesh Strategy:** Use a coarser mesh in large, uniform regions and a finer mesh near holes, edges, and fillets where stress concentrations may occur.

“Diving Board Example.x_t”



# Step 2: Set up the Material and Geometry

1. Open Ansys Workbench
2. Select and drag “static structural” on the left hand side into the open area
3. Open the “Engineering Data” by double clicking on it.
4. Click on “add new material” and name it “Diving Board”
5. On the left hand side click and drag density and isotropic Elasticity to the bottom box
6. Download the “Diving Board Example.x_t”
7. Right Click geometry, select import, and select the ‘Diving Board Example.x_t”







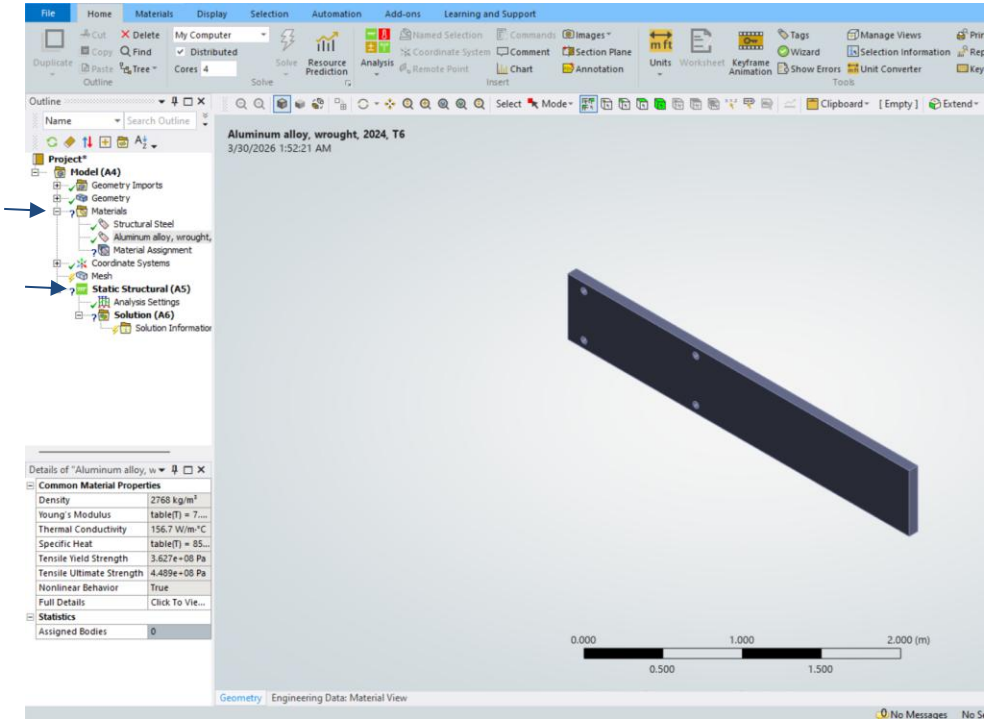
Set Density to 2.7 g/cm^3, Young's Modulus to 69000 MPa, and Poisson's Ratio to 0.33

Properties of Outline Row 3: Diving Board

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Property | Value | Unit | | |
| 2 | Material Field Variables | Table | | | |
| 3 | Density | 2.7 | g cm^-3 | | |
| 4 | Isotropic Elasticity | | | | |
| 5 | Derive from | Young's Modulus an... | | | |
| 6 | Young's Modulus | 0.069 | MPa | | |
| 7 | Poisson's Ratio | 0.33 | | | |
| 8 | Bulk Modulus | 67647 | Pa | | |
| 9 | Shear Modulus | 25940 | Pa | | |

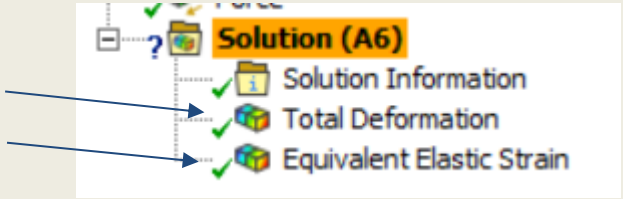
# Step 3: Adding Forces and Supports

1. Open the model by right clicking “Model” and opening “Ansys Mechanical”
2. Edit the current material of the model to be the “Diving Board/Fiberglass” configured before.
3. Right click “Static Structural” click “insert” choose “Fixed Support” and click the left side of the model
4. Right click “Static Structural” click “insert” choose “Force”
5. In the bottom left configure the magnitude to be 981N
6. Click the right most face of the model, and apply the “geometry”.
7. Click the top surface of the model and apply “direction”

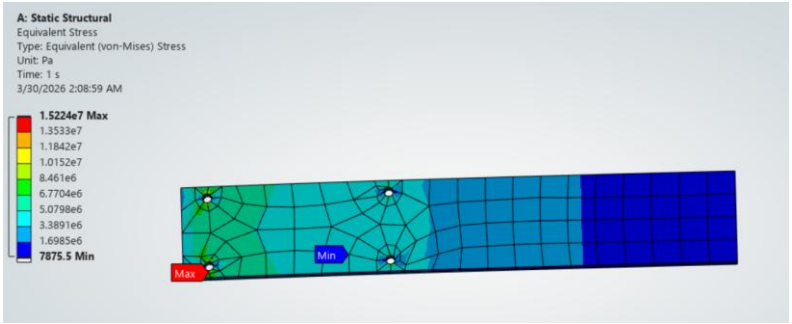


# Step 4: Solving

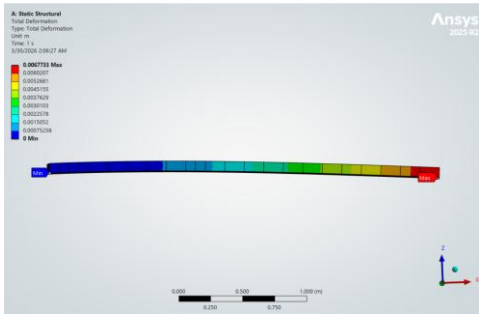
1. Right Click “Mesh”, click “Generate Mesh”
2. Right Click “Solution”, click “insert”, click “Deformation”, click “total deformation”.
3. Right Click “Solution”, click “insert”, click “Stress”, click “Equivalent (Von-Mises)”
4. Right click solution, and choose “Solve”
5. To Generated solutions, navigate to the feature tree and double click your desired solve.




Equivalent Stress Solve Overview:




Deformation Solve Overview:





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# Validation and Common Pitfalls

How to keep your first simulation from lying to you

# Best practices you can actually remember



### 1. Start simple

Get a basic model working before adding more realism. A clean simple model teaches you more than a messy complicated one.

### 2. Check assumptions

Ask whether the support, load, and material choices reflect the real situation closely enough.

### 3. Validate the trend

Use hand calculations and engineering intuition to make sure the response is believable.

# Common beginner mistakes

**Wrong units**

Probably the fastest way to get nonsense.

**Bad supports**

Too stiff or too loose can both break the model.

**Mesh too coarse**

Important gradients can get smeared out.

**Missing material data**

The solver cannot guess your intent.

**Believing color only**

A red region is not automatically failure.

**No sanity check**

Results without physics checks are risky.

# How to tell whether a simulation makes sense



**Ask Yourself:**

- Does the deformation shape match intuition?
- Do the highest stresses appear in a location that makes mechanical sense?
- Are the units consistent?
- Would a simple hand calculation predict the same trend?
- If the mesh is refined a bit, do the key outputs stay in the same ballpark?

**Validation Steps:**

1. The setup is realistic
2. The trend matches engineering intuition
3. The result is stable enough to be believable



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# Wrapping up

What you should remember!

# Key takeaways

- FEA is a numerical way to estimate how parts behave under real-world conditions.
- Ansys organizes the process clearly: geometry, materials, mesh, loads, solve, and results.
- The software can be powerful, but good answers still depend on good assumptions.
- Advanced tools like contact, thermal, modal, and optimization build on the same foundation, but will likely not currently be used much.

FEA does not replace engineering judgment. It gives engineering judgment something much better to work with.



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Next Up

# Week 7: FEA II



# “Testing Leads to Failure, and Failure leads to Understanding”

-Henry Petroski

Incubator